

Agent: host_agent

All special education needs school sites in Exeter have been found and plotted. Here’s what was done:

- Located the area polygon for Exeter.
- Searched for all school sites classified as "School For Special Needs" within Exeter.
- Found 7 such school sites, including:
- Ellen Tinkham School Further Education College
- West of England School and College
- Southbrook School
- Magdalen Court School
- Barley Lane School

A map has been generated showing all these special education needs school sites in Exeter, with each site displayed as a polygon and labeled with its name and area.

If you’d like to view the map or need more details about any of the schools, let me know!

Agent: planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is special education needs school sites, and we need to find as many entries as possible for these schools in Exeter.

output steps: ["Find Exeter 1 area", "Search for special education needs school sites in Exeter as many search results"]

Tool Call: send_message

- target: named_area
 - task_description: Find the area polygon for Exeter.
-

Agent: named_area_agent

The specific polygon for Exeter has been filtered and saved as:

- **Artifact Name:** filtered_exeter_area_polygon
 - **Description:** GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and settlement/city/town/suburban/village description.
-

Tool Call: call_os_ngd

- filters: None
- bbox: None
- polygon_or_point: True
- street_address: None
- filename: exeter_area_polygon

Result: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter., count of records fetched : 98.

Tool Call: send_message

- target: data_analysis_agent
 - task_description: Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon.'
-

Agent: data_analysis_agent

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons specifically representing Exeter
    # Use name1_text for 'Exeter', and also check for relevant descriptiongroup/description
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower().str.contains('exeter'))
        ) & (
            (gdf['descriptiongroup'].isin(['Settlement', 'Part Of Settlement'])) |
            (gdf['description'].isin(['City', 'Town', 'Suburban Area', 'Village']))
        )
    ]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and settlement/city/town/suburban/village criteria."
    return [
        summary,
        'filtered_exeter_area_polygon',
        'GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and settlement/city/town/suburban/village description.',
        filtered
    ]
```

Filtered to 1 polygon specifically representing Exeter using both the name and relevant settlement/city/town/suburban/village criteria. - Artifact name: filtered_exeter_area_polygon - Description: GeoDataFrame containing only the polygon(s) specifically representing Exeter, filtered by name and settlement/city/town/suburban/village description. - The artifact contains the filtered polygon data for Exeter.

Tool Call: send_message

- target: land_use_features
- task_description: Search for special education needs school sites within the artifact 'filtered_exeter_area_polygon.'

Agent: land_use_features_agent

The search for special education needs school sites within the 'filtered_exeter_area_polygon' has been completed. Here are the results:

- **Filtered Artifact Name:** filtered_special_education_needs_schools_exeter
- **Description:** GeoDataFrame containing only the school sites in Exeter identified as 'School For Special Needs' according to the 'oslandusetierb' column. All original columns and geometry are retained.

- **Number of Entries Found:** 7

If you need further assistance or details, feel free to ask!

Tool Call: call_os_ngd

- filters: ['School']
- bbox: filtered_exeter_area_polygon
- polygon_or_point: True
- street_address: False
- filename: special_education_needs_schools_exeter

Result: Multiple search results have been found in multiple datasets. Artifacts generated are : ['landuse_special_education_needs_schools_exeter'], Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for special education needs school sites within the artifact 'filtered_exeter_area_polygon'. using filters ['School']. Further Name filtering is available for this"], counts of records fetched are : [45].

Tool Call: send_message

- target: data_analysis_agent
- task_description: Filter the dataset 'landuse_special_education_needs_schools_exeter' to identify special education needs school sites. Save the filtered results as 'filtered_special_education_needs_schools_exeter'. Include filtering using the 'oslandusetierb' column.

Agent: data_analysis_agent

Tool Code Executor

```
def filter_special_education_needs_schools_exeter(data: list):  
    gdf = data[0]  
    # Filter for special education needs schools using 'oslandusetierb' column  
    filtered_gdf = gdf[gdf['oslandusetierb'].str.contains('Special Needs', case=False, na=False)]
```

```
summary = f"Filtered special education needs school sites: {len(filtered_gdf)} entries found."
artifact_name = 'filtered_special_education_needs_schools_exeter'
artifact_description = (
    "GeoDataFrame containing only the school sites in Exeter identified as 'School For Special Needs' "
    "according to the 'oslandusetierb' column. All original columns and geometry are retained."
)
return [summary, artifact_name, artifact_description, filtered_gdf]
```

Filtered special education needs school sites: 7 entries found. A new artifact named 'filtered_special_education_needs_schools_exeter' has been generated. This GeoDataFrame contains only the school sites in Exeter identified as 'School For Special Needs' according to the 'oslandusetierb' column, with all original columns and geometry retained.

Tool Call: send_message

- target: plotting_agent
- task_description: Plot the artifact 'filtered_special_education_needs_schools_exeter' to show all special education needs school sites in Exeter.

Agent: plotting_agent

Tool Code Executor

```
def plot_special_education_needs_schools_exeter(data: list):
    import folium
    import geopandas as gpd
    from shapely.geometry import mapping

    gdf = data[0].copy()
    # Ensure CRS is EPSG:4326 for folium
    if gdf.crs is not None and gdf.crs.to_epsg() != 4326:
        gdf = gdf.to_crs(epsg=4326)

    # Center map on the centroid of all schools
    centroid = gdf.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=12)

    # Add polygons for each school
    for idx, row in gdf.iterrows():
        folium.GeoJson(
```

```

        mapping(row['geometry']),
        name=row['name1_text'],
        tooltip=f"{row['name1_text']}<br>Area: {row['geometry_area_m2']:.0f} m²"
    ).add_to(m)

folium.LayerControl().add_to(m)
map_filename = 'special_education_needs_schools_exeter_map.html'
m.save(map_filename)

# Prepare summary
summary_rows = gdf[['name1_text', 'geometry_area_m2', 'status']].head(5)
summary = (
    f"Plotted {len(gdf)} special education needs school sites in Exeter. "
    f"First 5 schools: "
    f"\n" + summary_rows.to_string(index=False)
)
description = (
    "A folium map showing all special education needs school sites in Exeter. "
    "Each site is shown as a polygon with a tooltip for the school name and area."
)
return [summary, 'special_education_needs_schools_exeter_map', description, map_filename]

```

Plotted 7 special education needs school sites in Exeter. First 5 schools: - Ellen Tinkham School Further Education College (6776.62 m²) - West of England School and College (50703.99 m²) - Southbrook School (20679.40 m²) - Magdalen Court School (4602.29 m²) - Barley Lane School (18341.40 m²)

Artifact name: special_education_needs_schools_exeter_map

Description: A folium map showing all special education needs school sites in Exeter. Each site is shown as a polygon with a tooltip for the school name and area.

Map file: special_education_needs_schools_exeter_map.html